

EcoWeighing S.A.

1 Description

1.1 Introduction

After several IT students had driven to a recycling center and spent what felt like half a day in the queue there, they came up with the idea of making this process more efficient. To save a lot of time, they thought of small, self-sufficient drop-off points that could even be set up POPup-style for a short time if needed.

In addition, they also wanted to collect data and provide information about each customer's environmental footprint in order to positively influence their recycling behavior.

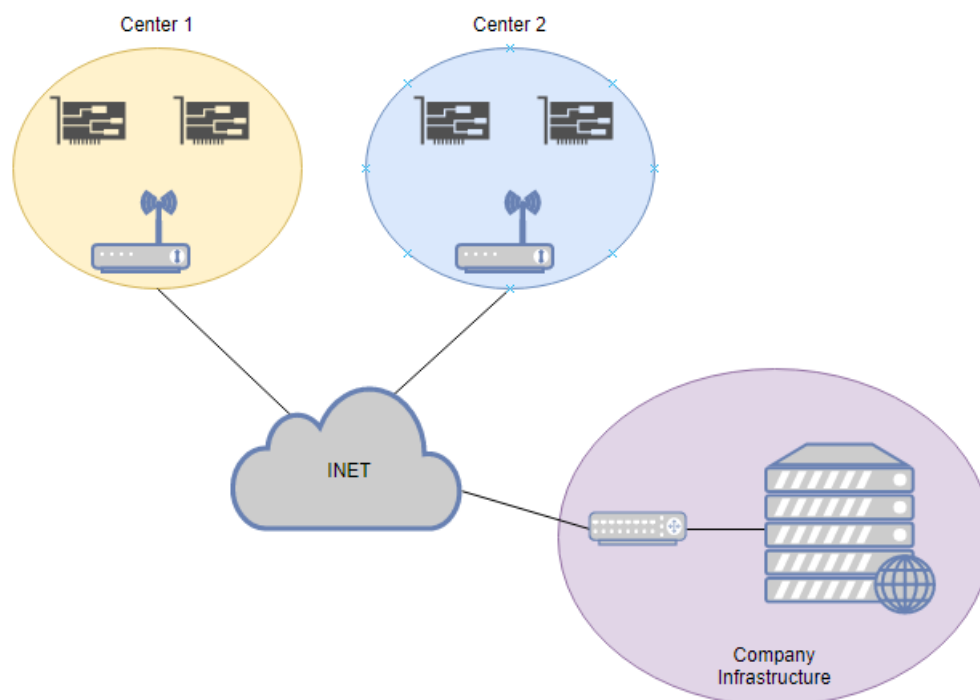
With this idea in mind, they founded the company EcoWeighing S.A. to produce their intelligent scales bundled with a software solution for the entire web-based management.

The scales are fully remote configurable, meaning that the waste type, station name and location can be customized to suit the customer's needs.

To make their company a success the founders decided to apply the fundamental steps of good project management and split the development and production of the scales into several parts (sprints). These parts consist, among other things, of the planning and construction of the measuring stations (scales) and the programming of the web-based software solution. The software provides user administration and the possibility to manage all the centers and scales.

Moreover, to illustrate and facilitate the uncomplicated integration into existing systems, a network simulation and a virtual machine with all software components (web server, database) are created.

1.2 General Structure



Sprints	Chapter
The model to be made in the workshop serves as a basis and contains the necessary electronics, a measuring element, display and a microcontroller.	Workshop
The website allows users to create an account, log in and get an overview of all past measurements.	Website
The database stores the data.	Database
The controller interacts with the end user, takes measurements, and interacts with the website.	Controller
The integration of these measuring stations into different network structures is simulated in Cisco Packet Tracer.	Network

2 Timeline

KW	Labo	Workshop (2h. each)	Special
38			Stage
39			
40			
41			
42			
43			
44	Congé de la Toussaint		
45	L1 - Database	Hand over components and material according to the material list. Soldering of Printed Circuit Boards	Read competences, program and assignment
46	L1 - Database		
47	L2 - Website	A1 - Soldering of Printed Circuit Boards	User story "website" must be planned
48	L2 - Website		
49	L2 - Website	A2 - Wiring of weighing unit onto Plexi Plate	
50	L2 - Website		
51	L2 - Website	A3 - PVC Plate 3e/a	
52	Vacances de Noël		
1			
2	L2 - Website		Interim presentation 1
3	L3 - Packet Tracer	A4 – Metal Weighing Plate Production	
4	L3 - Packet Tracer		
5	L4 - Micro controller		
6			
7	Congé de Carnaval		
8	L4 - Micro controller	A5 - Production of Metal Bracket	User story "server" must be planned
9	L5 - Web & Database server		
10	L5 - Web & Database server		
11	L5 - Web & Database server		
12	L6 - Micro controller integration	A6 - Assembly and commissioning of the model	User story "integration" must be planned
13	L6 - Micro controller integration		
14	Vacances de Pâques		
15			
16			Last Sprint & Interim Presentation 2
17			
18			
19			
20			
21			Final levy
22	Congé de la Pentecôte		
23			

3 Implementation

This chapter gives you an overview of the steps that must be performed to implement the project.

3.1 Workshop

In order to reduce the future CO2 footprint, efficient recycling scales must be produced for the various recycling centers.

Each recycling scale should contain a personalized electronic unit. This unit also includes the following modular boards: ESP32-CAM, OLED display, weighing unit. The programming of the controller needs to be realized by means of a second programming unit. Both boards can be used modularly.

The boards are to be installed on an appropriate model.

This model includes a PVC base plate with 2 side panels. These are installed at the front end. The upper end of the side panels is connected with a prefabricated Plexi-plate. The Plexi is equipped with a 1kg scale. The rear end of the base plate is an aluminum bracket, which carries the electronic unit.

On the weighing unit there is an aluminum bowl for the weight measurement of the various materials. The aluminum bowl can also be replaced by a self-made 3D printed bowl.

All components must be manufactured independently, on time and professionally. Power is supplied via an external power supply unit.

3.1.1 List of workshop user stories

Note: For all user stories all work steps must be fully documented.

Userstory 1 Workshop: Production of both printed circuit boards

The board of the ESP32 and programming board according to kit type must be assembled and soldered correctly. All work steps must be fully documented.

Userstory 2 Workshop: Production of the Plexi weighing unit

The prefabricated Plexi is to be equipped with the 1kg weighing unit and switch S1. The connection conductors of the switch S1 and the weighing unit to the main board are to be prepared with shrink tubing for further processing.

Userstory 3 Workshop: Production of the PVC base plate and the side panels

A PVC base plate including two side panels is to be produced according to the plan. The base plate is used to mount the metal bracket. The side panels are used to assemble the prefabricated Plexi including weighing unit.

Userstory 4 Workshop: Production of the weighing plate

A metal weighing plate to accommodate the recycled goods should be expertly manufactured according to the plan. The weighing plate is to be mounted on the weighing unit.

Userstory 5 Workshop: Production of the metal bracket

A metal bracket is to be manufactured according to the plan. The metal bracket is used to attach the ESP32 basic PCB.

Userstory 6 Workshop: Assembly and commissioning of the model

All manufactured components must be correctly connected to each other, and all surrounding components must be professionally wired together. It must be made sure that the components are correctly arranged. The finished model should be put into operation and all functions should be recorded in a test report. Any errors must be corrected.

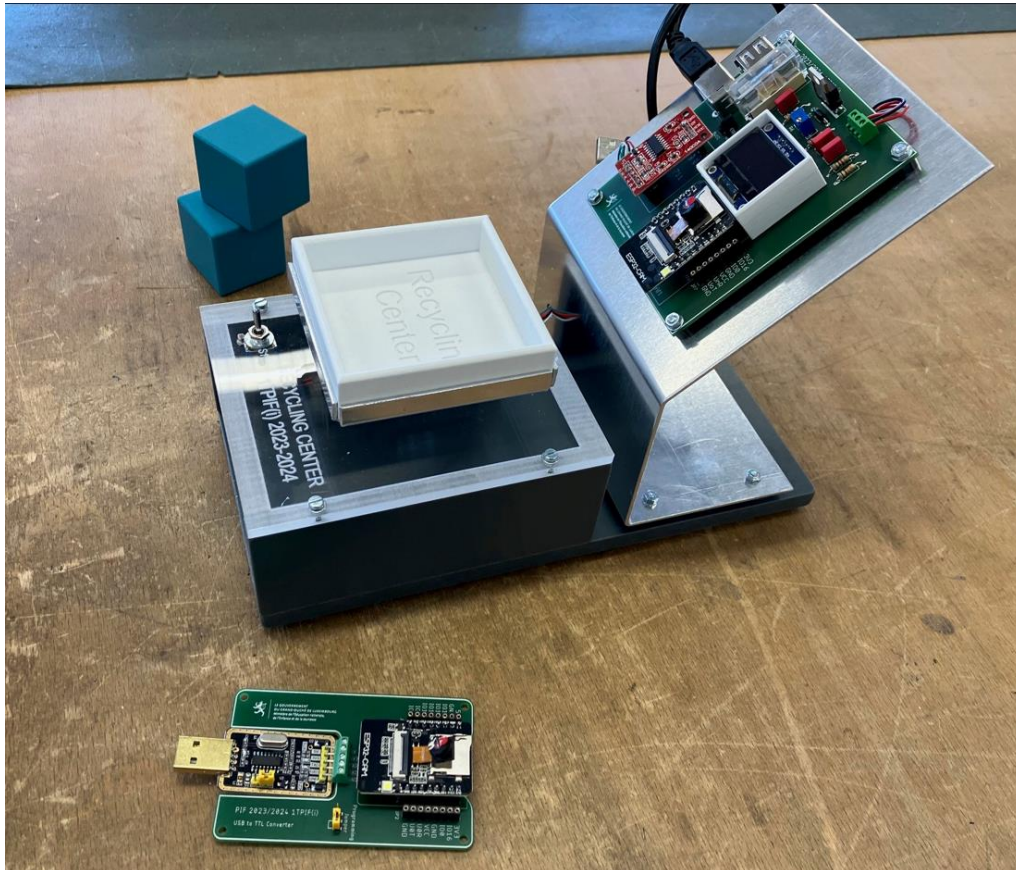


Figure 1 Recycling Center

3.2 Website

3.2.1 Introduction

To create a functional website, some key features must be implemented. The default page without being logged in must show a dynamically generated description (address, opening hours, etc.) based on the database entries of one random center among the ones in the database. There could be a drop-down menu to select other additional centers. Registered users must have the option of choosing a favorite center, which is then displayed on the default webpage.

You must be able to register, log in, and log out of the website. This is crucial to ensure that user data is secure and only accessible to authorized users.

In order to maintain the security, the user's password cannot be stored in plain text in the database. Instead, it must be hashed to protect it from potential hacks and data breaches. Additionally, the user's input, including their name, username, password, and email address, must be validated on the server side to prevent any malicious activity. The username must be unique, the email must be a valid email and the password must be strong.

Users must also be able to edit their profiles, all fields including their password can be updated by respecting the given constraints. This ensures that users have control over their own data and can keep it up to date as needed. The same validation rules used for the login should be applied before changing any data.

One unique feature that this website must have is the ability to generate a unique, random alphanumeric code (string) for each registered user. For security reasons, this code must be built as a new random alphanumeric string (rather than a simple counter), and without any reference to the user credentials. This code will be visible to the user after they log in and will be used to identify them at recycling centers.

The users are able to see an overview of the existing measurements in the database. For this the last 10 measurements, shall be visualized in a detailed table like shown below. A drop-down menu could be added to select the number of measurements to be displayed (for example 10, 20, 100, ...).

	Recycling Center	Date	Type	Weight (kg)
1	Luxembourg-City	05/05/2023	Plastic	15
2	Luxembourg-City	05/05/2023	Grass	30
...				
10	Luxembourg-City	21/04/2023	Grass	25

To provide users with a more comprehensive view of their activity on the site, a full billing history must be presented to each user. On each bill the details of all waste disposals and the amount that is due as well as the total sum for the corresponding day are displayed, like shown in the database part. This will give users more transparency and control over their usage and help them track their recycling efforts.

To ensure that the code is well-documented and easy to understand for anyone who may be working on the website in the future, clear comments must be written. These comments should include detailed explanations of what each part of the code does.

Finally, once the basic functionality of the site has been established, it's important to focus on aesthetics. Some styling must be applied to make the site more visually appealing and user-friendly.

3.2.2 Improvements (Additional functions)

To create an improved version of your website, there are several key features that should be implemented. First, to manage users effectively, there should be two types of users in the system: administrators and clients.

Only administrators should have the ability to add, edit, and delete recycling centers, waste types, and waste balances of any recycling center. Moreover, administrators should be able to delete existing users as well. For the best performance, the website could ultimately allow administrator users to fully edit all the data of other users. The administrator could also be able to manage centers.

If an administrator account has been created the view of the webpage should be adapted to its specific needs compared to the view of the regular client.

To improve performance, the validation of user input should also be done on the client side as well, to avoid unnecessary network traffic. This will help ensure that the site runs smoothly and efficiently.

Furthermore, the administrator could be provided with tools for statistical analysis of trash disposal (either by recycling center, trash type or users). Different statistical indicators could be used for that purpose (such as frequency, mean, standard deviation, etc.) and a graphical representation of the data.

Regular users could be able to view custom reports of their waste disposal (based on either recycling center or waste type).

To ensure that your code is modular and well-organized, repetition of code should be avoided, and functions should be extensively used both in client and server-side code. This will help ensure that your code is easy to maintain and that changes can be made quickly and efficiently.

To give users more control over their information, all users should have the ability to regenerate and replace their "identifier code". Additionally, users could be able to use a QR code creation library to generate the QR image and display it within the website.

Furthermore, the styling of the website should be improved with a responsive design to make it more visually appealing and user-friendly.

Additionally the validation functions written in PHP could be reused by the client-side script using asynchronous requests (AJAX). This ensures that user input is validated thoroughly and efficiently, reducing the risk of errors and security breaches.

By implementing these features, your website will become more secure, user-friendly and efficient. It will also provide a better experience for both clients and administrators alike.

3.3 Database

In this part you will design a database that can save the data required by the project.

The design will begin with a UML conceptual model of the database including all classes, associations, and cardinalities.

The resulting database will later be used as the backend to the recycling center website.

As many recycling centers will be distributed throughout the country, every center has its own code. Address and opening times must be saved in the database.

Only registered clients can bring their waste for recycling. It will be enough to save the username, name, email, password and unique random alphanumeric code(string) which is generated when the user registers for the first time. Every user has their own unique username which is to be used as primary key. Here is an example of a user entry:

Username	Name	Email	Type (opt.)	RandomCode	Password	Fav. Center
jschmit	Jos Schmit	jschmit@pt.lu	user	345fgt678	xxxxxxxxx	RCLUX
dmueller	Daniel Mueller	dmueller@admin.net	admin	fgt66781	xxxxxxxxx	

Every Recycling center must be recorded in the database with its full address and opening times. Every center has a unique identifier. Here an example of a recycling center entry:

Center code	Address	Postal code	Town	Open	Close
RCLUX	1, rue de Beggen	112345	Luxembourg	8:30	18:00

It must be possible that the user's favorite recycling center is recorded in the database.

Every measurement station where a user weighs its waste must also be recorded in the database. A station has a unique number and a description. A measurement station belongs to a specific recycling center. This must be traceable.e...

As the price the user needs to pay, depends on the weight and the type of waste, all types must be recorded in the database. Every waste type has a unique code by which it is identified. Here an example of a waste entry:

WasteCode	Description	Price
PLA	Plastic	0.2€
BIO	Kitchen Bio Waste	0.1€

When a user weighs its waste, it must be possible to save not only the weighted value, but also the type of waste, the measurement station, and the recycling center in which the action took place. Here is an example of such a measurement:

Recycling Centre: Luxembourg-City	Code: RCLUX			
User: Jos Schmit				
Measurement Station No 145	Measure No	Weight(kg)	Type	Date
Description: Hard plastic and glass materials	451	25	Plastic	05/05/2023
	452	30	Flat Glass	05/05/2023
Measurement Station No 146	4625	15	Kitchen bio waste	05/05/2023
Description: Biological	4626	30	Grass	05/05/2023

The user's measurements of the day will then result in an invoice where the prices for the individual measurements as well as the total price will be listed. Below an example of such an invoice:

Invoice No: 2657		Date: 06/05/2023		
User: Jos Schmit				
Recycling Centre: Luxembourg-City				
Summary	Measure No	Weight(kg)	Type	Price
	451	25	Plastic	5.0€
	452	30	Flat Glass	7.0€
	4625	15	Kitchen bio waste	1.5€
	4626	30	Grass	4.0€
TOTAL:		17.5€		

The operators of the recycling center would also like to know if an invoice was paid or not.

3.4 Web- and Database Servers

To create a website that uses a database in the background you need a web- and a database server. In this project you will set up a virtual machine that runs these servers.

The latest Ubuntu Server LTS version, an Apache webserver, MySQL, and PHP stack as well as the OpenSSH server have to be installed on this server.

The virtual machine should have 2 network interfaces. Each interface gets its own static address. The first interface is used to access the website with the usual ports, the second is used for the exchange with the microcontroller, and port 8080 should be used for this. You should implement this function with the help of Virtualhosts.

To save time, the website should already be developed locally and only migrated to the server later once it had been set up. To facilitate the migration, phpMyAdmin should also be installed and configured on the web server.

3.5 Controller

As company employee, you will need to configure and set up the weighing station on the site of the recycling center. To do that the ESP32-CAM must be flashed with the provided software. The Arduino IDE is used for this, and libraries must be installed first.

Each weighing station has a unique ID that must be defined in the code. Moreover, the connection to the server must be configured in the code by specifying the following:

- the SSID,
- the password of the WLAN,
- the host or domain name of the server,
- and the file to access.

In addition to that, for test purposes a program should be written which allows to use the scale stand alone, meaning without integration of the QR-code reader and the webserver integration.

The weighing station is set up with two predefined parameters in the code to scale the weighing station:

- `scale_scale (= a)`
- `scale_offset (= b)`.

In order to improve the accuracy of the weighing station, the employee should adjust these parameters in the provided source code. For this purpose, standardized test weights can be used to obtain reference measures. On this basis, new values for the two parameters can be calculated by using a linear equation.

In the following the procedure of the waste disposal using the scale and the QR-code reader is explained. First the client enters the recycling facility and must show the QR-code to gain access to the site and to identify themselves. The web server uses the customer ID and a database to check whether the client is authorized to dispose of their waste. The web server then responds with either “**access granted**” or “**access denied**”. If the customer has been granted access, he can place the waste on the weighing station and turn on the switch to start the measurement. Three comma-separated values are then sent to the web server and processed by a PHP-script:

- the weight of the waste,
- the ID of the weighing station,
- and the client ID based on the QR-code.

If no value is measured, the user is asked to place the waste on the balance again, by enabling the switch the measurement is started.

Note for each new measurement, the client first needs to turn on the switch and then scan the QR code again.

3.6 Network

The aim is to illustrate to potential clients the integration into their network. Therefore, a fictitious company network is to be set up with the help of Packet Tracer. To be able to respond as authentically and quickly as possible to individual customer needs, your simulation will consist of several parts:

- The company headquarters with its server room
- The network of an ISP
- Several centers
- The Internet with some web servers (to simulate a solution with an external server if necessary).

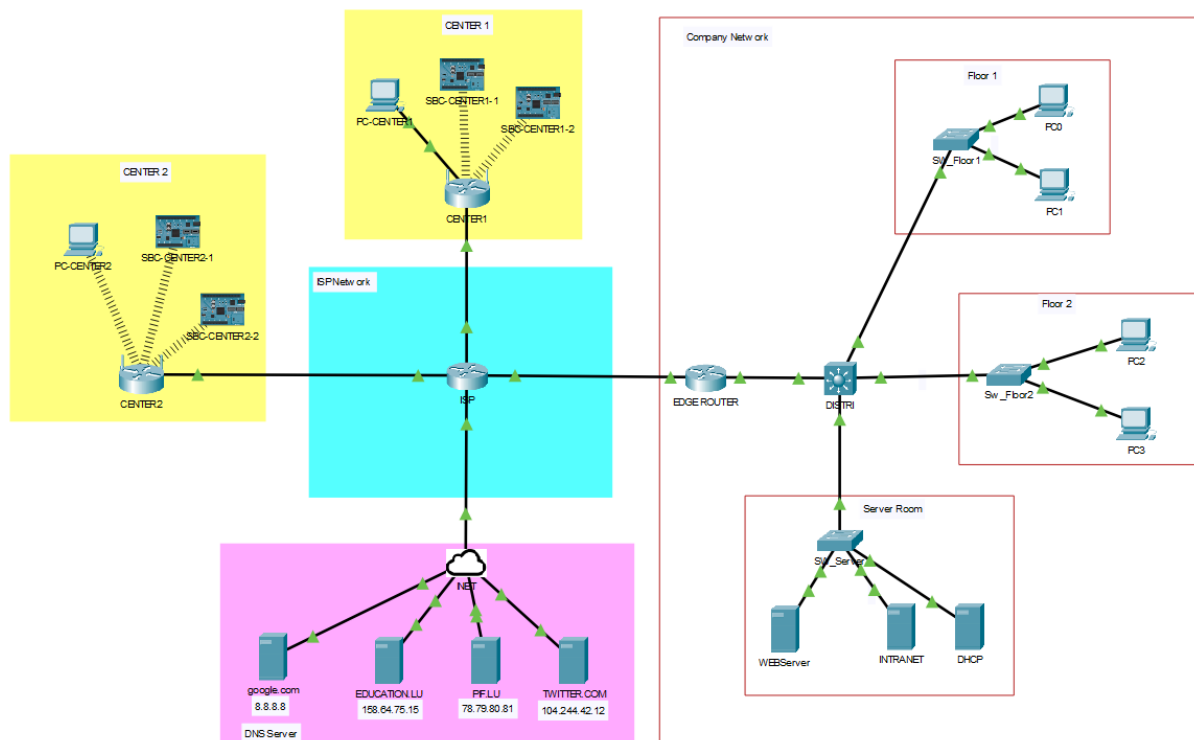


Figure 1

As a network technician, you are assigned by the project manager to create such a simulation. This is done in Cisco Packet Tracer (version 8.1 or higher). A sketch created by the project manager (Figure 1) should serve as an implementation aid!

For performance, security and cost reasons, company networks today are always divided into individual VLANs. The VLANs relevant for the simulation are listed in Table 1.

VLAN	Name	Network	Description
10	USER1	192.168.10.0/24	User of a department 1
20	USER2	192.168.20.0/24	User of a department 2
99	DMZ	192.168.99.x/yy	Server, for one or more WEB servers to which the determined data are sent.
100	SERVER	192.168.100.0/27	Server, for all other company servers
199	MGMT	192.168.99.x/yy	Management network for all wired network devices
50	WIFI	192.168.50.0/24	Opt. WLAN access for the company network
299	MGMT_WIFI	192.168.99.x/yy	Opt. Ap Communication with WLC

Table 1

As shown in the Table 1 the management networks and the DMZ are not yet fully defined. You should define them according to Table 2 with the help of VLSM.

IP address range to be divided	Network	Required IP addresses
192.168.99.0/24	VLAN 99	10
	VLAN 199	50
	VLAN 299	25

Table 2

Furthermore, the project manager wants easy-to-remember IP addresses for all networks connected to the ISP network (ISP router). For this you must reach into the bag of tricks, as you do not want to plan the networks unnecessarily large. In Table 3 you will find the defined network addresses together with the network.

Device	IP address	Network
Company Edge Router	11.11.11.11	11.11.11.8/29
Centre 1 Router	22.22.22.22	22.22.22.16/28
Centre 2 Router	33.33.33.33	33.33.33.32/29

Table 3

Note the following guidelines:

- All the devices and connections in Figure 1 must be considered.
- A colleague had already started this task. His work, a table of all interfaces, networks, IP addresses, is available to you. And must be completed.
- Home routers are used for the “Centre” networks, they are the best solution for this, as all the necessary functions are integrated, such as a switch, a router, NAT, DHCP, WLAN, etc. These are configured with the help of a GUI.
- The L2 switches model are all Cisco Catalyst 2960.
- The L3 switch model is Cisco Catalyst 3650 (used as L2 Switch in the must part)
- All routers are model Cisco PT_Empty except the Edge Router (ISR 4321).
- The network configuration (IP address, netmask, default gateway, DNS server) of the hosts in VLAN 10 and VLAN 20 are done via the DHCP server shown in the network diagram. Note that a special configuration is required to forward the DHCP-specific traffic to all / foreign subnetworks!
- The network configuration of all other hosts (servers) in the company network, as well as the routers, is static.
- In all networks, the first host address is assigned to the router.
- The Edge Router must be configured with NAT functionalities. The router should also allow the forwarding from the HTTP and HTTPS port to the DMZ web server. This server receives the data from the stations and serves as a customer portal.
- Create an ACL which limits the traffic for devices belonging to USER2 VLAN. No Access to the Server with the customer data should be possible.
- A dedicated and secured access to network devices management Interface by a separated MGMT VLAN have to be implemented.

After you have completed your simulation, you must save the current configurations (running-config) of all network devices (switches and routers) on the network devices in Packet Tracer (startup-config) and add them to your documentation. For better readability leave only the relevant lines.

Your documentation also includes the fully completed table with all assigned IP addresses based on the spreadsheet you got at the beginning.

3.6.1 Optional Elements

After some customer demonstrations, it turns out that not all the required components are sufficiently implemented. The following possible extensions, some are shown in Figure 2, should be considered:

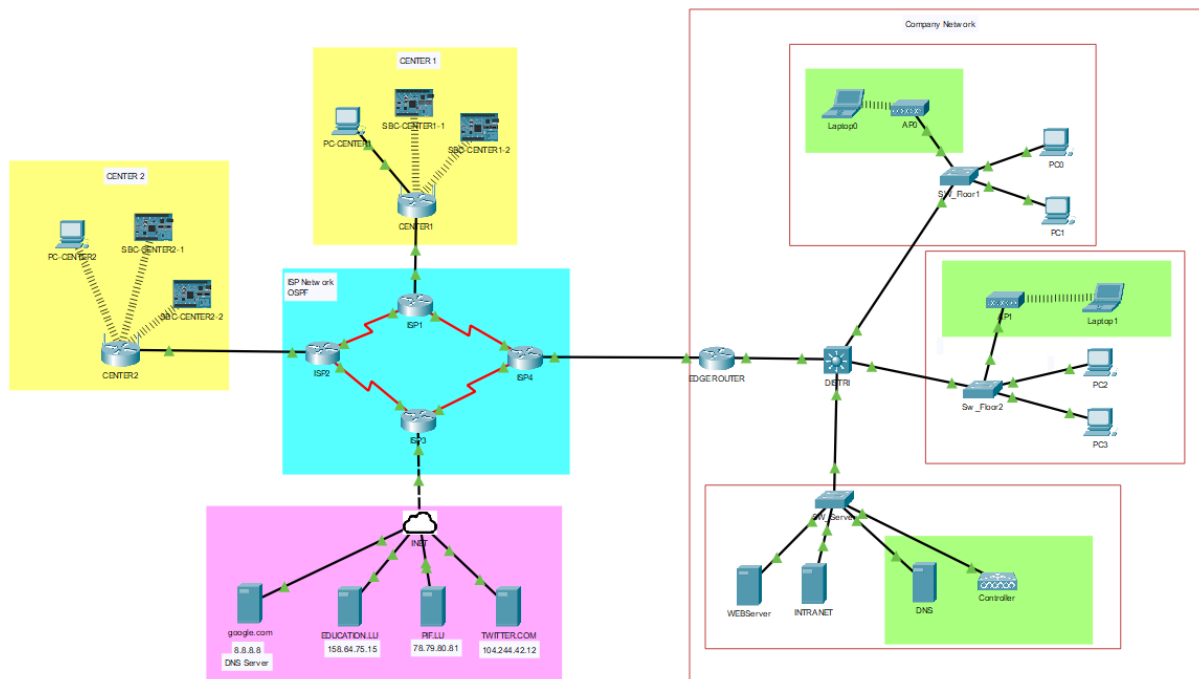


Figure 2

Add a WLAN service to the company network:

- Bring in a WLAN controller and several APs
- The WLAN controller and all APs should be manageable via VLAN 299 (WLAN_MGMT).
- At least one SSID (WIFI) should be available in encrypted form. For security reasons, this will be a separate subnet (VLAN50).
- Access from this network should be purely to the Internet!

The company network should have its own DNS server:

- There should be one entry each for the DMZ server and the intranet server.
- All other requests should be forwarded to the External DNS server accordingly.

The ISP network is to become somewhat more complex, and is to be realized with the help of a dynamic routing profile:

- 4 routers are to be connected using OSPF and serial interfaces.

2 tier network architecture with L3 switch:

- Configure the network so that the L3 switch is responsible for internal routing (L3 Inter VLAN Routing). Only access to the ISP network and NAT are done via the edge router.
- Migrate and upgrade the ACL to the L3 switch, and restrict the access for the WIFI VLAN
- Configure the DHCP functions on the L3 switch, for all VLAN except MGMT_WIFI. Leave this either on the external DHCP Server or use the integrated DHCP on the WIFI Controller

Enable secure system administration at least on the edge router with SSH. In case of public access to the Edge router management interface at least the traffic is encrypted and passwords not easily readable.

4 Material List

QTY:	Part Number	Description
Electronic Hardware		
1	BOU 3266W-1-101	Spindeltrimmer, 12 Gänge, 100 Ohm, linear
1	1N 4007	Gleichrichterdiode, 1000 V, 1 A, DO-41
1	1/4W 360	Widerstand, 360 Ohm
1	1/4W 240	Widerstand, 240 Ohm
1	1/4W 5,1K	Widerstand, 5,1 kOhm
1	1/4W 51K	Widerstand, 51 kOhm
3	MKS2-100 100N	MKS2 PET-Kondensator, 100 nF
1	LM 317-220	Spannungsregler, einstellbar, 1,2 ... 37 V, TO-220
1	STV PTF/15	Sicherungshalter für 5 x 20 mm, 250 V, 6,3 A
1	STV BS140	Sicherungshalter für 5 x 20 mm, 250 V, 6,3 A, transparent
2	MPE 087-1-002	Stiftleisten 2,54 mm, 1X02, gerade
1	USB AW	USB-Einbaubuchse, Serie A, gew., Printmontage
1	USB BW	USB-Einbaubuchse, Serie B, gew., Printmontage
1	PHC 1725669	Leiterplattenklemme 3-polig, RM 2,54 mm
8	MPE 115-1-008	Präz.-Buchsenleisten 2,54 mm, 1X08, gerade
		Zur Info: Die 1x10 Buchsenleiste muss auf 1x8 gekürzt werden.
1	PHC 1725672	Leiterplattenklemme, 4-polig, RM 2,54 mm
1	MPE 115-1-006	Präz.-Buchsenleisten 2,54 mm, 1X06, gerade
2	MPE 115-1-004	Präz.-Buchsenleisten 2,54 mm, 1X04, gerade
1	MPE 087-1-003	Stiftleisten 2,54 mm, 1X03, gerade
2	JUMPER 2,54GL SW	Kurzschlussbrücke, schw. m. Griffflasche.

		Zur Info: Benötigt wird nur 1 Jumper an der 3,3V Versorgungsspannung. 1 Jumper ist Reserve.
1	DEBO HX711-01	Entwicklerboards - A/D Wandler - Waage, 1 kg
1	DEBO CAM ESP32	Entwicklerboards - ESP32-Kamera, 2MP, 25°
1	DEBO OLED2 0.96	Entwicklerboards - Display, 0,96", OLED-Display, SSD1306
1	MS 500D	Kippschalter 6A-125VAC, 1x Ein-Ein
1	OTB 8013930	USB-Ladegerät, 5 V, 1000 mA, schwarz, USB
1	DEBO USB TTL	Raspberry Pi -USB-TTL-Schnittstellenwandler, CH340G
1	GC 2510-05OFS	USB A > B Stecker, 1 m
12	VT DK 5MM M3	Distanzhülsen, Kunststoff, M3 5mm
4	VT DK 5MM M4	Distanzhülsen, Kunststoff, M4, 5mm
2	HT 151-02244	Klebesockel 19x19mm MB3APT-A schwarz
1	PHC 1725685	Leiterplattenklemme, 5-polig, RM 2,54 mm
1	MPE 087-1-004	Stiftleisten 2,54 mm, 1X04, gerade
		Zur Info: Wird benötigt beim A/D Wandler DEBO HX711-01
2	PCB	2 Leiterplatten (Main + Programming)
1	ESKA 521.517	Feinsicherung 5x20mm, mittelträge (m), 1A
4	3M SJ5012 BLACK	Gummifüße, 12,7 x 3,5 mm
2	GC KAB-10S25	Kabelbinder, UV-beständig, 100 mm, schwarz, 100er-Pack
	HT 308-30300	Schrumpfschlauch, 3:1 Ø 3,0 mm, schwarz
	HT 308-30600	Schrumpfschlauch, 3:1 Ø 6 mm, schwarz
	HT 308-30150	Schrumpfschlauch, 3:1 Ø 1,5 mm, schwarz
	LITZE RT	Kupferlitze isoliert, 100 m, 1 x 0,14 mm², rot
	LITZE SW	Kupferlitze isoliert, 100 m, 1 x 0,14 mm², schwarz
Metal		
1		sheet metal 290x120 t=2mm eloxé
1		sheet metal 100x100 t=2mm eloxé

Plexi		
1		Plexi Plate 140x120 t=5mm
PVC		
1		PVC Plate 1 250x120 t=10mm grey
1		PVC Plate 2 140x50 t=10mm grey
1		PVC Plate 3 140x50 t=10mm grey
Screw Hardware		
8	M3x20	screw (Sheet metal to PVC + Plexi to PVC)
4	M3x16	screw (PVC to PVC)
20	M3	washer
8	M3	nut
4	M4x16	screw (1kg Weighing scale to Plexi and Alu bowl)
4	M4	washer